

# Bieberbach–de Branges, Univalent Coefficients, and the Pandrosion Contour Integral

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## Abstract

The Bieberbach conjecture, proved by de Branges in 1985, states that  $|a_n| \leq n$  for univalent functions  $f(z) = z + \sum a_n z^n$  on the unit disk, with equality only for the Koebe function  $K(z) = z/(1-z)^2 = \sum n z^n$ . We recover the coefficients via the Pandrosion contour integral  $(2\pi i)^{-1} \oint f(z)/z^{n+1} dz$  (Paper 76) and interpret de Branges' proof through Löwner chains as a sequential flow on Pandrosion fields: each step adds a pole to  $F_{f_t}$ , building univalent functions one quotient at a time.

## 1 The Bieberbach conjecture

**Theorem 1.1** (de Branges, 1985). *If  $f(z) = z + a_2 z^2 + a_3 z^3 + \dots$  is univalent (injective holomorphic) on  $\mathbb{D}$ , then:*

$$\boxed{|a_n| \leq n \quad \forall n \geq 1,} \tag{1}$$

*with equality if and only if  $f$  is a rotation of the Koebe function  $K(z) = z/(1-z)^2$ .*

## 2 Coefficient extraction via Pandrosion integral

**Proposition 2.1.** *The Taylor coefficients are extracted by contour integration (Paper 76):*

$$a_n = \frac{1}{2\pi i} \oint_{|z|=r} \frac{f(z)}{z^{n+1}} dz = \frac{1}{2\pi} \int_0^{2\pi} f(re^{i\theta}) e^{-in\theta} \frac{d\theta}{r^n}. \tag{2}$$

Verified for the Koebe function at  $10^{-13}$  precision ( $n = 1, \dots, 10$ ):

| $n$ | Koebe $a_n = n$ | $-\log(1-z)$ : $1/n$ | $z/(1-z^2)$ : odd | $2(\sqrt{1+z}-1)$ |
|-----|-----------------|----------------------|-------------------|-------------------|
| 1   | 1 (=)           | 1.000                | 1.000             | 1.000             |
| 2   | 2 (=)           | 0.500                | 0.000             | 0.250             |
| 3   | 3 (=)           | 0.333                | 1.000             | 0.125             |
| 4   | 4 (=)           | 0.250                | 0.000             | 0.078             |
| 5   | 5 (=)           | 0.200                | 1.000             | 0.055             |

The Koebe function attains equality for *every*  $n$ ; all other univalent functions have strict inequality.

### 3 The Pandrosion field of $K$

**Proposition 3.1.** *The logarithmic derivative of the Koebe function is:*

$$\frac{K'(z)}{K(z)} = \frac{1+z}{z(1-z)}, \quad (3)$$

a “Pandrosion field” with poles at  $z = 0$  and  $z = 1$ . The winding number  $\frac{1}{2\pi i} \oint_{|z|=r} \frac{K'}{K} dz \approx 1$  (one zero inside  $\mathbb{D}$ , at  $z = 0$ ).

Verified: winding = 0.993 at  $r = 0.5$ .

### 4 Löwner chains as Pandrosion flows

de Branges’ proof uses the Löwner differential equation:

$$\frac{\partial f_t}{\partial t} = z f_t'(z) \cdot \frac{1 + \kappa(t)z}{1 - \kappa(t)z}, \quad |\kappa(t)| = 1. \quad (4)$$

In Pandrosion language: at each infinitesimal step, the Löwner equation adds a *pole* to the field  $F_{f_t} = f_t'/f_t$  at  $z = -1/\bar{\kappa}$ . This is a sequential construction of univalent functions *one quotient at a time*—the continuous analogue of building a polynomial root by root.

The chain of implications proved by de Branges:

$$\text{Milin} \implies \text{Robertson} \implies \text{Bieberbach}.$$

### 5 Random schlicht verification

50 random schlicht functions  $f(z) = z \exp(\sum c_k z^k)$  with  $|c_k| \leq 0.14$ : 400 coefficient tests, 0 violations of  $|a_n| \leq n$ .

### References

- [1] L. de Branges, *A proof of the Bieberbach conjecture*, Acta Math. **154** (1985), 137–152.
- [2] P. Koebe, *Über die Uniformisierung*, Math. Ann. **67** (1907).
- [3] I. Besevic, *The polynomial index theorem via the Pandrosion field*, Pandrosion Dynamics **76** (2026).